

Chapter 2

Simplified forms of the N-S equations

In Chapter 1, we have presented and discussed the Navier-Stokes equations, which are the most general form of the equations of fluid dynamics. In this Chapter, we shall take a look at various simplified forms of the equations.

These simplified forms differ from the *Navier-Stokes equations* in that they miss selected terms on the account that, for specific flow configurations, these terms are negligible and do not change the solution to the equations appreciably.

There are important advantages to solving simplified forms of the equations when appropriate. Firstly, less terms means that the equations may be discretized more readily. Secondly, depending on the terms that are neglected, the equations may change mathematical character (see Chapter ??), thereby becoming more suitable for specialized and efficient solution methods.

Of course, neglecting terms in the governing equations poses some risks too. These simplifications may or may not be warranted, depending on the flow configuration itself. In other words, removing terms from the N-S equations is equivalent to neglecting certain physical processes. But, how does one know *a priori*, which terms and physical processes can be safely neglected? In reality, experience guides the choice of the most suitable form of the Navier-Stokes equations. Moreover, once a solution to the simplified equations is available, the missing terms can be evaluated *a posteriori* and their magnitude compared against the terms that have been retained.

There are complexities associated with the simplification of the N-S equa-

tions. In particular, a viable simplification must yield a set of equations that allow for as many boundary conditions as we wish to apply and that retain physically meaningful dependencies of the flow properties from spatial coordinates. The simplification of the Navier-Stokes equations by trimming terms is not unlike the Japanese art of Bonsai. The *trimmed* form of the Navier-Stokes equations must be viable and useful. Too little trimming defies the purpose of seeking a simplification. Too much trimming may result in a system of PDEs without a solution or with a grossly inaccurate and useless solution.

Detailed discussions on the simplified forms of the N-S equations presented in this Chapter are found in Batchelor [1967].

2.1 Equations for incompressible and isothermal flows

This simplified form of the Navier-Stokes equations is, by far, the most widely used set of equations that describe fluid flow. They are obtained from the N-S equations assuming that the density is constant ($\rho = \text{const}$) and that variations in the temperature of the fluid, if any, are negligible, so that the viscosity of the fluid is also constant to a very good approximation ($\mu = \text{const}$).

Examples of flows that are well-described by these simplified equations are flows involving liquids without temperature inhomogeneities and slowly moving gases. Even when the fluid is a gas, a *slow moving gas*, i.e. a gas in a flow configuration where $\text{Ma} = |\mathbf{u}|/a \leq 0.3$, behaves as if it were incompressible. Here, Ma is the *Mach number* and a is the speed of sound in the gas. In other words, the condition $\text{Ma} = |\mathbf{u}|/a \leq 0.3$ ensures that changes in velocity are associated with minor changes in pressure that, in turn, induce only minor changes in the density of the gas. Thus, density can be safely assumed to be constant as its variations are negligible and do not influence the velocity field.

Under the assumptions above, the N-S equations become

$$\nabla \cdot \mathbf{u} = 0, \tag{2.1}$$

$$\frac{D\mathbf{u}}{Dt} = -\nabla\pi + \nu\nabla^2\mathbf{u} + \mathbf{b}, \tag{2.2}$$

2.1. EQUATIONS FOR INCOMPRESSIBLE AND ISOTHERMAL FLOWS²³

where $\nu = \mu/\rho$ is the kinematic viscosity (SI units are $\text{m}^2 \text{s}^{-1}$) and we have defined the *normalized pressure* $\pi = p/\rho$. The symbol ∇^2 is the *Laplacian* operator, resulting in second order spatial derivatives. In this simplified form of the momentum equation, the Laplacian is applied to the velocity vector, component by component.

The equations above are four scalar equations (or one scalar equation and one vector equation) in the four unknown $\mathbf{u} = (u_1, u_2, u_3)$ and π . Note that the energy equation is no longer included.

The first equation, which is a simplified form of continuity under the assumption that $\rho = \text{const}$, indicates that, for incompressible flows, the only admissible velocity vector fields are those with a *zero divergence* or *solenoidal vector fields*.

The second equation is a simplified form of the momentum equation under the assumption that both $\rho = \text{const}$ and $\mu = \text{const}$. Notice that density no longer plays any role, rather the unknown is a *normalized pressure* π and the only coefficient that appears in the momentum equation is the kinematic viscosity ν .

An important consequence of this situation is that flows with fluids of different density, but equal kinematic viscosity ν , will admit the same velocity and normalized pressure (π), if the flow geometry and boundary conditions are the same.

Of course, if the two flows have the same π field, they will have different pressures fields since $p = \rho\pi$, with the fluid with a higher density displaying pressure fields of greater magnitude.

2.1.1 The pressure equation for incompressible isothermal flows

It is possible to replace the scalar continuity equation $\nabla \cdot \mathbf{u} = 0$ with an equation for pressure π . The pressure equation is obtained by taking the divergence of the momentum equation and manipulating the results. One obtains

$$\nabla^2 \pi = -\nabla \cdot (\mathbf{u} \cdot \nabla \mathbf{u}) + \nabla \cdot \mathbf{b} = S(\mathbf{u}, \mathbf{b}). \quad (2.3)$$

The equations above is an elliptic equation for π . The right hand side of the equation is a function of the velocity field and, if applicable, the body force.

Notice that the PDE does *not* include time derivatives. In other words, the pressure π is determined *instantaneously* by the velocity and body force fields and the boundary conditions.

It is customary to replace the continuity equation with the pressure equation above. Solution methods for the system of equations (Eq. (2.3) and Eq. (2.2)) are available and popular.

2.1.2 The vorticity equation for incompressible isothermal flows

It is possible to replace the momentum equation (Eq. (2.2)) with an equation for vorticity. Recall that vorticity is the *curl* of the velocity vector

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}, \quad (2.4)$$

where $\nabla \times$ indicates the curl. Thus, the vorticity is a vector that contains linear combinations of partial spatial derivatives of the velocity components.

Obtaining a PDE for $\boldsymbol{\omega}$ is straightforward, albeit somewhat tedious. The derivation starts by taking the curl of the momentum equation.

Suffice to say that, because the curl of the gradient of a scalar field is always equal to zero, regardless of the field, the gradient of pressure drops off the vorticity equation, i.e. $\nabla \times \nabla \pi = 0$. In fact, this is the reason why a PDE for vorticity may be preferred to a PDE for velocity, as it results in eliminating pressure from the PDE.

The vorticity equation for incompressible isothermal flows reads

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + \mathbf{u} \cdot \nabla \boldsymbol{\omega} = \frac{D\boldsymbol{\omega}}{Dt} = \boldsymbol{\omega} \cdot \nabla \mathbf{u} + \nu \nabla^2 \boldsymbol{\omega} + \nabla \times \mathbf{b}. \quad (2.5)$$

The first term on the r.h.s. of Eq. (2.5) is called the *vortex stretching and tilting term*, acting as a source of vorticity. The second term on the r.h.s. is akin to the term $\nabla^2 \mathbf{u}$ in the momentum equation and indicates that vorticity is transported by *diffusive processes* in the presence of viscosity. The third term on the r.h.s reflects the fact that vorticity can be generated only by a body force that features a non-zero curl.

Notice that the vortex stretching and tilting term features the product of vorticity and derivatives of the velocity components. Thus, one needs the interaction of vorticity *and* velocity gradients to change vorticity.

2.2 Inviscid flow (Euler flow)

This limiting form of the Navier-Stokes equations is obtained by neglecting all terms that feature viscosity. The ensuing equations are referred to as the *Euler equations* in honor of the Swiss scientist Leonhard Euler¹.

They read

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2.6)$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = \nabla p + \rho \mathbf{b} \quad (2.7)$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot (\rho \mathbf{u} H) = \rho \mathbf{b} \cdot \mathbf{u} + s \quad (2.8)$$

These are 5 equations in the 5 unknowns ρ , $\mathbf{u} = (u_1, u_2, u_3)$, and E . Notice that pressure is obtained from the density ρ and total energy E given the definition of total energy and an equation of state (EOS) specific to the fluid under consideration.

The absence of the terms that feature viscosity or diffusivity has very important implications with regard to our ability to solve the system of PDEs. Recall that the viscous and diffusive terms involve second order derivatives. Dropping these second order derivatives makes the Euler equations *hyperbolic PDEs*, which are rather straightforward to solve. Thus, you have an example of a simplified form of the Navier-Stokes which is simpler to solve numerically than the full equations.

An obvious and reasonable question is whether the Euler equations are an adequate physical model for fluid flow, considering that all fluids have a non zero viscosity. Well, *the answer depends on the flow configuration and the question being asked*.

The answer to this question is, in fact, much more complex than it may appear. It may help to realize that the Bernoulli equation, named after the Swiss mathematician Johann Bernoulli² – who was Euler's doctoral advisor

¹Leonhard Euler (15 April 1707 - 18 September 1783) was a Swiss mathematician, physicist, astronomer, logician and engineer, who made important and influential discoveries in many branches of mathematics.

²Johann Bernoulli (6 August 1667 – 1 January 1748) was a Swiss mathematician and was one of the many prominent mathematicians in the Bernoulli family. He is known for his contributions to infinitesimal calculus and educating Leonhard Euler in the pupil's youth.

incidentally, is in fact an integral form of the momentum equation (Eq. (2.7)). So, you probably know that the Bernoulli equation is used frequently to characterize the relation between velocity and pressure. The Bernoulli equation is widely used in engineering and found to be very useful for design. It follows that the Euler equations must be useful too.

Yet, neither the Bernoulli equation, not the Euler equations allow to describe viscous stresses and physical processes that rely on molecular motion, which often times are important at walls and surfaces of technical devices or at the interfaces of streams with very dissimilar flow properties (i.e. in the presence of large gradients in velocity, temperature, density, etc.). Unfortunately, estimates of the drag forces and heat fluxes and mass fluxes to the wall of a device are often *precisely* what is needed for design. Notable examples include the viscous drag on an airplane's wings or the walls of a pipe, or the heat flux to the heat shields of a space vehicle during re-entry. The Euler equations will not answer those questions.

However, if one is interested in the characterization of the flow properties *away from walls and interfaces*, then the Euler equations are an accurate and dependable physical model. How far away from walls and interfaces one needs to be depends on extent of regions of the flow where processes due to viscosity and diffusivity exert an influence. We call these regions *boundary layers*. We shall provide a small introduction to boundary layers in Section 2.4.

2.3 Stokes flow equations

Starting from the equations for isothermal and incompressible flows, if the flow is characterized by a very small Reynolds number, the convective terms in the momentum equation may be safely neglected. If the time derivative of the velocity is also negligible, one lets $D\mathbf{u}/Dt = 0$ in the momentum equation:

$$\nabla \cdot \mathbf{u} = 0 \tag{2.9}$$

$$\nu \nabla^2 \mathbf{u} - \nabla \pi + \mathbf{b} = 0 \tag{2.10}$$

These equations are called the Stokes limit of the N-S equations and the flow configurations that these PDEs apply to are often called *Stokes flow*. Of course these are named after George Gabriel Stokes of the Navier-Stokes fame.

Examples of Stokes flow are *creeping flows* (small sizes and velocities), flows in coating processes (high viscosity), micro-devices (very small sizes), and bearings (small gaps and high viscosity). Some flows of small insects and organisms are also well described as Stokes flows.

The most important feature of the equations in the Stokes flow limit is that the *non-linearity* of the full Navier-Stokes equations has disappeared on the account that the non-linear convective term has been removed from the momentum equation. Furthermore, there are *mirror symmetries* in the flow that emerge naturally from the PDEs. Some of these symmetries are illustrated well by the visualization of streamlines, whereby it is not possible to indicate the direction in which the fluid is flowing.

This simplification is very consequential, making the PDEs for the velocity $\mathbf{u}$ and pressure π linear, *elliptic*, and mirror symmetric. As you can imagine, the numerical solution of the Stokes equations is rather straightforward on the account of their linearity.

2.4 Boundary layer equations

A boundary layer is a small region of the flow where viscous and diffusive processes are important and affect the flow field. But what are exactly viscous and diffusive processes?

Viscous and diffusive processes are physical processes that influence momentum and energy because the viscosity and diffusivity of the fluids are not zero. Looking at the full Navier-Stokes equations, viscous processes are those described by terms containing the viscosity of the fluid and diffusive processes are those described by terms containing the viscosity, diffusivity, or thermal conductivity of the fluid.

As you recall, these terms always contain *gradients* of the quantities for which the conservation laws are written. Thus, when the gradients of these fields are large in magnitude, such as inside boundary layers, these terms play a very important role. Examples include the velocity and temperature (energy) fields. The study of velocity boundary layers was pioneered by the German engineer Ludwig Prandtl³.

³Ludwig Prandtl (4 February 1875 – 15 August 1953) was a German engineer. He was a pioneer in the development of the mathematical treatment of aerodynamics, which have come to form the basis of the applied science of aeronautical engineering.

The most conspicuous example of boundary layers is the velocity boundary layer that forms on a wing (or a plane surface) as air flows around it. The boundary layer takes the form of a thin region of space, *coating the wing* and confined between the surface of the wing and the *far field*. As one moves outwards along a direction normal to the wing's surface, the boundary layer is traversed.

The surface of the wing, aligned approximately parallel to the direction of the incoming flow, imposes a preferential direction (the *streamwise* direction) and a direction perpendicular to it (the *normal* or *crosswise* direction). If the flow shows three-dimensional features and dependencies, the third direction orthogonal to the streamwise and crosswise directions is considered also (the *spanwise* direction).

Under the assumption that the flow has a preferential streamwise direction, it is possible to simplify the full Navier-Stokes equations because certain terms are negligible. In particular, one expects that gradients in velocity will be much greater in the direction normal to the surface (crosswise) than along the surface (streamwise). Furthermore, the following simplification often assumes that the flow is steady and all unsteady terms are neglected also.

We do not go into the details of the derivation, but simply present the so called *boundary layer equations*, which we write *explicitly* in Cartesian coordinates for the sake of simplicity. In what follows x_1 represents the coordinate aligned with the streamwise direction and x_2 is the coordinate normal to it. Similarly, u_1 is the velocity component in the streamwise direction and u_2 the component in the direction normal to it (crosswise component). We only present the equations for a two-dimensional boundary layer flow.

$$\frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} = 0, \quad (2.11)$$

$$u_1 \frac{\partial u_1}{\partial x_1} + u_2 \frac{\partial u_1}{\partial x_2} = \nu \frac{\partial^2 u_1}{\partial x_2^2} - \frac{\partial \pi}{\partial x_1}, \quad (2.12)$$

$$\frac{\partial \pi}{\partial x_2} = 0. \quad (2.13)$$

Equation (2.11) is the continuity equation for an incompressible isothermal flow and is identical to the continuity equation we have seen as part of the Navier-Stokes equations. Equations (2.12) and (2.13) are the momentum

equations for the streamwise (u_1) and crosswise (u_2) components of velocity, respectively.

Due to the simplifications related to the directionality of the flow, the momentum equation for u_2 (Eq. (2.13)) reduces to a simple statement on the lack of dependence of the normalized pressure from the normal coordinate x_2 . In other words, Eq. (2.13) implies that $\pi = \pi(x_1)$ and the normalized pressure does not vary *across the boundary layer*, only *along the surface oriented in the streamwise direction*.

Then, all we are left with are two equations (Eq. (2.1) and Eq. (2.12)) in three unknowns $u_1 = u_1(x_1, x_2)$, $u_2 = u_2(x_1, x_2)$ and $\pi = \pi(x_1)$. Clearly, this is a challenging outcome of the simplifications we have made to the full Navier-Stokes equations. How can one solve a system of PDEs with more unknowns than equations? Of course, one cannot and the normalized pressure π *must be prescribed before one can actually solve the equations*.

But where does one obtain the normalized pressure field π around an object with arbitrary shape? Suitable manners of obtaining π include potential flow theory supplemented by Bernoulli's equation or the solution of the Euler equations in the region surrounding the object. Because the Euler equations describe the flow accurately *away and outside of the boundary layers* and because the normalized pressure π is constant across the boundary layer, i.e. the pressure outside and inside the boundary layer is the same, the Euler equations provide a very good approximation to the evolution of pressure on the surface of an object immersed in a moving fluid.

During the early history of CFD, the approach of solving the steady Euler equations around a wing, followed by the solution of the two-dimensional boundary layer equations has proven very valuable in the design of airplanes.

In closing, we remark that the two-dimensional steady boundary layer equations shown above are *parabolic* in character and are often solved with what are called *marching methods*.

2.5 Summary and take-away messages

- The Navier-Stokes equations may be simplified under specific assumptions. A simplified form of the N-S equations is applicable to flow configurations whereby the assumptions are satisfied and suitable.
- Any simplified form of the N-S equations offers advantages in terms of

being more tractable and simpler to solve.

- The isothermal and constant density form of the N-S equations is arguably the most widely used form of the N-S. It is applicable to describe the flow of liquids and slow moving gases.
- The Euler equations are obtained from the Navier-Stokes equations upon neglecting all terms that contain viscosity, diffusivity, and thermal conductivity. The Euler equations are non-linear and *hyperbolic* in character. They describe fluid flow well *away from surfaces* and outside of *boundary layers*.
- The Stokes equations are obtained from the N-S equations under the assumption that the material derivative of velocity is negligible. The ensuing equations are *elliptic* in character, linear, and mirror symmetric.
- The boundary layer equations describe the flow inside thin regions where gradients are most important in a direction normal to surfaces and walls, compared to the direction of the flow. The boundary layer equations are *parabolic* and solved with *marching methods* along the streamwise direction, i.e. the main flow direction.

Bibliography

G K Batchelor. *An introduction to fluid dynamics*. Cambridge University Press, New York, 1 edition, 1967.